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05/24/26

CS 499 Milestone Two: Enhancement One — Software Design and Engineering

Artifact Description

The artifact I selected is my CS 360 Project Three Inventory App. I created this project in Android Studio using Java and XML for my Mobile Architecture and Programming course. The app allows users to log in, manage inventory records, store data locally, and use SMS notification features.

Justification for Inclusion in the ePortfolio

I selected this artifact because it shows my skills in Android development, user interface design, event handling, input validation, and software structure. This project is a good ePortfolio artifact because it shows how I can take a basic working application and improve it into a cleaner and more maintainable product.

To complete this software function enhancement project, we implemented two core code modifications: we added the `ValidationUtils` helper class to separate verification logic and achieve code decoupling, and refactored `InventoryActivity` by splitting its add, delete, and update methods to eliminate code redundancy, which comprehensively improved the code's readability, reusability, and maintainability.

Course Outcomes

This enhancement supports Course Outcome 2 because I am explaining my technical work clearly in this narrative and ePortfolio. It supports Course Outcome 3 because I evaluated the

existing app design and improved it using software design principles. It supports Course Outcome 4 because I used Android Studio, Java, XML, and software engineering practices to improve a working application. It also supports Course Outcome 5 because better validation helps reduce errors and supports safer handling of user input.

Reflection

While enhancing this artifact, I learned that working code is not always the same as well-designed code. The original app worked, but some logic was placed directly inside activity files, which made the code harder to maintain. By creating `ValidationUtils.java` and organizing inventory actions into separate methods, I improved separation of concerns and made the project cleaner.

I encountered a challenge related to the attribution of logical code during an Android development project. I had previously fallen into a common industry pitfall of writing all code directly into an activity. This optimization effort allowed me to master modular design, clean code, and software maintainability, which supports my career goals in software development and cybersecurity.

Conclusion

This project milestone has helped me improve my software design capabilities for the CS 360 inventory application. The optimized application features a clear architecture, comprehensive verification, strong readability, and secure input processing, which enables me to professionally showcase this work in my electronic portfolio.

Reference, National Institute of Standards and Technology. (2024). *The NIST cybersecurity framework (CSF) 2.0*. U.S. Department of Commerce. <https://doi.org/10.6028/NIST.CSWP.29>